

A Positive Action Plateau from Classical Velocity Memory

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Abstract

A finite reversible classical velocity process determines a positive limiting action coefficient through its integrated velocity correlation. We define the coefficient operationally as mass times displacement variance divided by observation duration, derive its exact spectral formula and bounds, and compute a two-velocity example. Finite speed forces the short-duration coefficient to vanish while a positive long-duration plateau survives. This separates microscopic resolution from observation time and identifies the correlation law that a universal action-scale proposal must explain.

1 The observable and its classical premises

The candidate action field is an ensemble observable

$$\mathbf{h}_\Delta(t) = \frac{m}{\Delta} \text{Var}[X(t + \Delta) - X(t)], \quad m, \Delta > 0. \quad (1)$$

Here X is position on the line, t is physical time and Δ is an observation duration. Its units are mass times length squared per time, hence action. For Gaussian increments with variance $\kappa\Delta/m$, this normalization yields $\mathbf{h}_\Delta = \kappa$, matching the earlier free-kernel model. For stationary velocity statistics the observable is independent of t , and we write $\mathbf{h}(\Delta)$.

The present premises are classical stochastic ones: a finite set of velocities, ordinary probabilities, finite transition rates, a stationary ensemble and detailed balance. Their mathematical consequences give a positive transport scale. The project's stronger target adds a physical origin for these premises, universality across systems, and a derivation identifying the scale with $\hbar$. We reserve $h = 2\pi\hbar$ for Planck's constants and use $\mathbf{h}$ for the candidate observable. An independent dynamical field would require an additional state variable and its equation of motion.

2 Finite speed and the correlation formula

The observation duration determines which part of the velocity dynamics is resolved. In particular, bounded velocity gives a sharp elementary constraint.

Proposition 1 (Microscopic limit). *If X has absolutely continuous paths with $|\dot{X}| \leq u < c$ almost everywhere, almost surely, then*

$$0 \leq \mathbf{h}_\Delta(t) \leq mu^2\Delta \longrightarrow 0 \quad (\Delta \downarrow 0).$$

Proof. The displacement has absolute value at most $u\Delta$. Its variance is at most its second moment, which is at most $u^2\Delta^2$. Divide by Δ/m . $\square$

For a real centered stationary velocity V with finite second moment, set $X(t) = \int_0^t V(s) ds$ and $C(s) = \mathbb{E}[V(s)V(0)]$. Fubini's theorem on each finite square is justified by Cauchy–Schwarz and stationarity. Symmetry of the covariance gives

$$h(\Delta) = 2m \int_0^\Delta (1 - s/\Delta) C(s) ds. \quad (2)$$

If $C \in L^1(0, \infty)$, dominated convergence yields

$$H_* := \lim_{\Delta \rightarrow \infty} h(\Delta) = 2m \int_0^\infty C(s) ds = 2mD. \quad (3)$$

This is the Green–Kubo diffusion coefficient D , with an action normalization [4, (1.3), Proposition 2.1]. The finite-state assumptions below make its strict positivity and convergence explicit.

3 A positive plateau from finite reversible dynamics

Let J_t be a stationary irreducible continuous-time Markov chain on $\{1, \dots, k\}$, $k \geq 2$, with invariant probabilities $\pi_i > 0$ and generator Q . Thus $Q_{ij} \geq 0$ for $i \neq j$, $Q\mathbf{1} = 0$ and $\pi_i Q_{ij} = \pi_j Q_{ji}$. All rates are finite. Assign velocities v_i with

$$\sum_i \pi_i v_i = 0, \quad \sigma_v^2 := \sum_i \pi_i v_i^2 > 0, \quad |v_i| \leq u < c.$$

Set $V(t) = v_{J_t}$ and integrate to obtain X . The Markov state is (X, J) ; the velocity label retains the information required for future motion. Paths are piecewise linear, with finitely many velocity jumps on compact intervals almost surely. These jumps represent impulsive interactions rather than a smooth prescribed force law.

Use $\langle f, g \rangle_\pi = \sum_i \pi_i f_i g_i$. On the mean-zero subspace, $-Q$ is positive definite: detailed balance gives

$$\langle f, -Qf \rangle_\pi = \frac{1}{2} \sum_{i,j} \pi_i Q_{ij} (f_j - f_i)^2,$$

and irreducibility makes its nullspace consist of constants. Let its positive eigenvalues be γ_j , with an orthonormal eigenbasis e_j , and write $v = \sum_{j=1}^{k-1} a_j e_j$.

Proposition 2 (Classical action plateau). *Under these premises,*

$$h(\Delta) = 2m \sum_{j=1}^{k-1} \frac{a_j^2}{\gamma_j} \left[1 - \frac{1 - e^{-\gamma_j \Delta}}{\gamma_j \Delta} \right], \quad (4)$$

$$H_* = 2m \langle v, (-Q)^{-1} v \rangle_\pi = 2m \sum_{j=1}^{k-1} \frac{a_j^2}{\gamma_j} > 0. \quad (5)$$

The inverse acts on mean-zero functions. Writing $\gamma_{\min} = \min_j \gamma_j$ and $\gamma_{\max} = \max_j \gamma_j$ gives

$$\frac{2m\sigma_v^2}{\gamma_{\max}} \leq H_* \leq \frac{2m\sigma_v^2}{\gamma_{\min}}, \quad 0 \leq 1 - \frac{h(\Delta)}{H_*} \leq \frac{1}{\gamma_{\min} \Delta}. \quad (6)$$

Moreover $h(\Delta)$ increases strictly from zero to H_ ; in particular $h(\Delta) \geq H_*/2$ for $\Delta \geq 2/\gamma_{\min}$.*

Proof. The transition semigroup gives $C(s) = \langle v, e^{sQ} v \rangle_\pi = \sum_j a_j^2 e^{-\gamma_j s}$. Inserting this into (2) and integrating each exponential proves (4). At least one a_j is nonzero, so the limit is strictly positive. The identity $\sum_j a_j^2 = \sigma_v^2$ gives the two spectral bounds. Each relative correction is bounded above by $1/(\gamma_{\min} \Delta)$, which gives the convergence estimate. Finally the scalar function $f(y) = 1 - (1 - e^{-y})/y$ satisfies $f'(y) = [1 - (1 + y)e^{-y}]/y^2 > 0$ for $y > 0$, since $e^y > 1 + y$. Its endpoint limits are zero and one. Positive weighted summation proves strict monotonicity and the stated duration-dependent bound. $\square$

This is a finite-state spectral specialization of the generator Poisson representation in [4, §2]. The bounds concern the observable (1) and fixed model parameters. Within the same axiom class, replacing Q by rQ for $r > 0$ changes H_* to H_*/r . Thus a uniform lower bound over models requires control of the transition rates as well as nonzero velocity variance. The spectral gap of the velocity generator has inverse-time units; H_* has action units.

4 Two velocities: the full time dependence

Choose velocities $\pm u$, $0 < u < c$, each with stationary probability $1/2$, and a reversal rate $\lambda > 0$:

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \lambda & -\lambda \end{pmatrix}, \quad V(t) = u\sigma_0(-1)^{N_t}.$$

Here N_t is a Poisson counting process of rate λ , independent of the uniform sign σ_0 . Its generating function gives $\mathbb{E}[(-1)^{N_s}] = e^{-2\lambda s}$, hence

$$C(s) = u^2 e^{-2\lambda s}, \quad h(\Delta) = \frac{mu^2}{\lambda} \left[1 - \frac{1 - e^{-2\lambda\Delta}}{2\lambda\Delta} \right]. \quad (7)$$

Therefore

$$h(\Delta) \sim mu^2 \Delta \quad (\Delta \downarrow 0), \quad H_* = \frac{mu^2}{\lambda} > 0 \quad (\Delta \rightarrow \infty).$$

The correlation time is $1/(2\lambda)$, so the limiting action is twice mass times velocity variance times correlation time. A particle starting at $X(0) = 0$ has support inside $[-u\Delta, u\Delta]$ at time Δ ; the no-reversal events give atoms at both endpoints. The model uses probability measures, including these atoms, rather than assuming a Gaussian density at every duration.

The velocity-resolved position measures $p_{\pm}(A, t) = \Pr\{X(t) \in A, V(t) = \pm u\}$ satisfy, distributionally,

$$\partial_t p_+ = -u\partial_x p_+ - \lambda p_+ + \lambda p_-, \quad \partial_t p_- = u\partial_x p_- + \lambda p_+ - \lambda p_-.$$

Adding and subtracting these equations and eliminating their difference gives the telegraph equation for $p = p_+ + p_-$,

$$\partial_t^2 p + 2\lambda\partial_t p = u^2\partial_x^2 p.$$

This is the persistent-flight construction associated with the classical telegraph process [1, pp. 1–2, (1.1)–(1.3)].

5 Which limit keeps the action?

Microscopic resolution and diffusion scaling give different values of the same observable. Fix $a > 0$ with diffusion units and take the formal family $u_{\lambda}^2 = a\lambda$. For every fixed $\Delta > 0$,

$$h_{\lambda}(\Delta) = ma \left[1 - \frac{1 - e^{-2\lambda\Delta}}{2\lambda\Delta} \right] \rightarrow ma.$$

Consequently the iterated limits are

$$\lim_{\Delta \downarrow 0} \lim_{\lambda \rightarrow \infty} h_{\lambda}(\Delta) = ma, \quad \lim_{\lambda \rightarrow \infty} \lim_{\Delta \downarrow 0} h_{\lambda}(\Delta) = 0. \quad (8)$$

This is an exact statement about second moments. The formal scaling sends $u_{\lambda} \rightarrow \infty$ and therefore leaves any fixed physical speed ceiling. At fixed $u < c$ and fixed λ , the positive plateau instead occurs for observation durations long compared with the correlation time.

6 Axiom audit and the universality question

Each premise has a visible mathematical role:

Classical premise	Consequence in the proof
Stationary centered ensemble	Correlation depends on elapsed time; mean drift is removed
Finite irreducible transition dynamics	Inverse generator exists on mean-zero functions
Detailed balance	Real orthogonal velocity modes with positive decay rates
Nonzero velocity variance	At least one mode contributes to the positive plateau
Finite speed	Short-duration action coefficient tends to zero
Mass and transition rates	Set the numerical value and units of the plateau

These premises establish positivity for each model in the class. Independent composition permits particles with different transition rates, so universality adds a further dynamical requirement. For the two-velocity family, a common limiting value H_* across masses requires

$$\lambda_m = \frac{mu_m^2}{H_*}. \quad (9)$$

Equation (9) states the rate law to explain. It is not supplied by independent composition. Likewise, the present effective model specifies impulsive reversals externally; a closed mechanical realization must account for their momentum exchange and correlation law.

Two reconstruction sources sharpen this requirement. Nelson’s stated Brownian hypothesis supplies $D = \hbar/(2m)$ at the outset [3, abstract]. Hall and Reginatto instead add momentum fluctuations whose strength scales inversely with position uncertainty; their ensemble coefficient C is identified with $(\hbar/2)^2$ [2, pp. 5, 8]. These are concrete comparison premises for the origin of a shared scale.

The next research step is to test whether a mechanical realization can enforce the mass-weighted correlation integral in (3) as a shared invariant. That would advance the scale-selection target beyond a family of independent transport coefficients. A separate phase/composition argument must then identify the common constant with $\hbar$.

References

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